Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. Write each final answer in its box. Only what is inside a box is graded.

Part A — Vocabulary (8 pts)

Write the letter of the best description in the box next to each term.

1. Index ☐
2. Radicand ☐
3. Principal root ☐
4. Conjugate ☐
5. Extraneous solution ☐
6. Composition ☐
7. Inverse function ☐
8. Anchor point ☐

(A) Everything written underneath the radical symbol; when the index is even, it may not be negative.

(B) The operation that feeds one function's output into another function's input, written $(f \circ g)(x)$.

(C) The small number in the crook of the radical symbol that says which power is being undone.

(D) The point (h, k) that positions a transformed radical graph — an endpoint when the index is even, a turning point when it is odd.

(E) A candidate that satisfies the powered-up equation but fails when it is tested in the original.

(F) The non-negative value the radical symbol by itself returns; a $\pm$ must be written in by hand.

(G) The function that reverses another's inputs and outputs; its graph is the mirror image over the line $y = x$.

(H) A binomial with the same two terms but the opposite middle sign; multiplying by it turns an irrational denominator rational.

Part B — Multiple Choice (12 pts)

Write the letter of the best answer in its box.

1. Written with a rational exponent, $\sqrt[3]{x^5}$ is ☐ (A) $x^{5/3}$ (B) $x^{3/5}$ (C) x^{15} (D) $x^{-3/5}$

2. The domain of $f(x) = \sqrt{7 - 2x}$ is ☐ (A) $x \geq \frac{7}{2}$ (B) $x \geq -\frac{7}{2}$ (C) $x \leq \frac{7}{2}$ (D) all real numbers

3. The graph of $y = \sqrt{3x - 12} + 5$ begins at the point ☐ (A) $(12, 5)$ (B) $(4, -5)$ (C) $(-4, 5)$ (D) $(4, 5)$

4. In simplest form, $\sqrt{75} + \sqrt{12}$ is ☐ (A) $\sqrt{87}$ (B) $7\sqrt{3}$ (C) $7\sqrt{6}$ (D) $10\sqrt{3}$

5. Which equation has *no real solution*? **Letter:** ☐

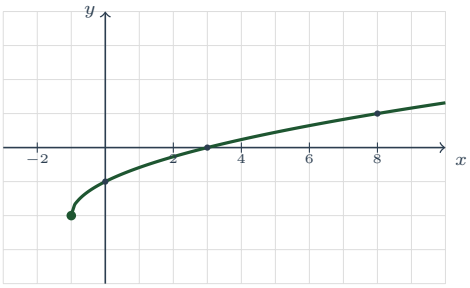
(A) $\sqrt[3]{x + 1} = -4$ (B) $\sqrt{x - 1} = 7$ (C) $\sqrt{x + 6} = -2$ (D) $\sqrt{x} = 0$

6. If $f(x) = \sqrt{x}$ and $g(x) = x + 9$, then $(f \circ g)(x) =$ ☐ (A) $\sqrt{x} + 9$ (B) $\sqrt{9x}$ (C) $x\sqrt{x} + 9\sqrt{x}$ (D) $\sqrt{x + 9}$

Part C — Short Answer & Computation (40 pts)

Show your work. Assume every variable is non-negative unless told otherwise.

1. **Read the graph.** The radical function f is graphed below; each gridline is one unit and the solid dot is the endpoint.



(a) Coordinates of the endpoint: ☐

(b) Domain (interval notation): ☐; range: ☐

(c) x-intercept: ☐; y-intercept: ☐

(d) Interval(s) where f is increasing: ☐

(e) Does f have an absolute maximum or an absolute minimum? Give it and say where it occurs.

(f) End behavior: as $x \to \infty$, $f(x) \to$; as $x \to -\infty$,

(g) Interval(s) where $f(x) < 0$:

2. **Radicals and rational exponents.** Rewrite or evaluate. Leave no negative exponents.

(a) Write $\sqrt[4]{x^3}$ with a rational exponent.

(b) Write $y^{5/2}$ as a radical.

(c) $27^{2/3} =$ (d) $16^{-1/2} =$

(e) Write $x^{1/2} \cdot x^{1/3}$ as a single radical. **Answer:**

3. **Simplify. Add or subtract where you can.**

(a) $\sqrt{72}$ (b) $\sqrt[3]{54}$ **(a)** **(b)**

(c) $\sqrt{20x^3}$ (d) $5\sqrt{12} - \sqrt{27}$ **(c)** **(d)**

4. **Multiply, then rationalize.** Every answer must be in simplest form.

(a) $\sqrt{6} \cdot \sqrt{15}$ (b) $\left(2 + \sqrt{5}\right)\left(2 - \sqrt{5}\right)$ **(a)** **(b)**

(c) $\dfrac{6}{\sqrt{3}}$ (d) $\dfrac{4}{3 - \sqrt{5}}$ **(c)** **(d)**

5. **Transformations.** Let $g(x) = -2\sqrt{x+3} + 4$.

(a) List every transformation that takes $y = \sqrt{x}$ to g.

(b) Anchor point: ____

(c) Domain: ____; range: ____

(d) Is the anchor an absolute maximum or an absolute minimum? ____

(e) A different graph of the form $y = \sqrt{x-h} + k$ begins at $(5, -2)$. Write its equation. ____

(f) For $p(x) = \sqrt[3]{x-1} + 2$, give the anchor point ____ and the domain ____. In one sentence, say why that domain is not like g's.

6. **Solve. Test every candidate in the original equation.**

(a) $\sqrt{2x+3} = x$ **Answer:** ____

(b) $\sqrt[3]{x-4} = -2$ **Answer:** ____

(c) $\sqrt{3x-2} + 4 = 1$ **Answer:** ____

(d) $x^{2/3} = 25$ **Answer:** [ ]

7. **Composition.** Let $f(x) = \sqrt{x}$ and $g(x) = x - 6$.

   (a) $(f \circ g)(x) =$ [ ], domain: [ ]

   (b) $(g \circ f)(x) =$ [ ], domain: [ ]

   (c) Are these the same function? Answer using *both* the rule and the domain.

   [ ]

   (d) $(f \circ g)(10) =$ [ ]

   (e) For $p(x) = 2x + 1$ and $q(x) = x^2 - 3$, find $(p \circ q)(-2)$. [ ]

8. **Inverses.**

   (a) $f(x) = 4x - 7$. Find $f^{-1}(x)$. [ ]

   (b) Verify by composition — *both* orders — that your answer really is the inverse.

   [ ]

   (c) $g(x) = \sqrt{x + 2}$. Find $g^{-1}(x)$ and state its domain. $g^{-1}(x) =$ [ ] **domain:** [ ]

   (d) The point $(7, 3)$ lies on the graph of g. What point must lie on the graph of g^{-1}? [ ]

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \sqrt{x-1} + 2$. **(a)** Give the anchor point, the domain, and the range, both in interval notation. **(b)** Find the x-intercept and the y-intercept, or explain why one does not exist. **(c)** List the transformations that take $y = \sqrt{x}$ to f. **(d)** Find $f^{-1}(x)$ and state its domain, saying exactly where that restriction comes from. **(e)** Justify with composition — both orders — that f and f^{-1} are inverses. **(f)** A classmate writes $y = (x-2)^2 + 1$ with *no* restriction and calls it the inverse. Explain why that is not the inverse of f.

2. **Modeling.** Ignoring friction, a roller-coaster car that starts from rest and drops h feet reaches the bottom at a speed of

$$v(h) = 8\sqrt{h} \text{ feet per second.}$$

(a) State the domain that makes sense here, and connect it to the rule about what an even index will accept. **(b)** Find $v(9)$ and $v(36)$, with units. What does quadrupling the drop do to the speed, and which feature of $\sqrt{\cdot}$ is responsible? **(c)** A car reaches 56 feet per second at the bottom. Find the height of the drop algebraically, and check your answer. **(d)** Write the inverse, h as a function of v, and say what it tells an engineer that $v(h)$ does not. **(e)** Part (c) required squaring both sides. Explain why there is no danger of an extraneous solution in this problem.